Chapter-6
Sets and Relations

Sets and Relations:

Set Operations, Representation and Properties of Relations, Equivalence Relations, Partially Ordering.

What is a SET

- Set are the fundamental discrete structures on which all the discrete structures are built. Sets are used to group objects together, formally speaking

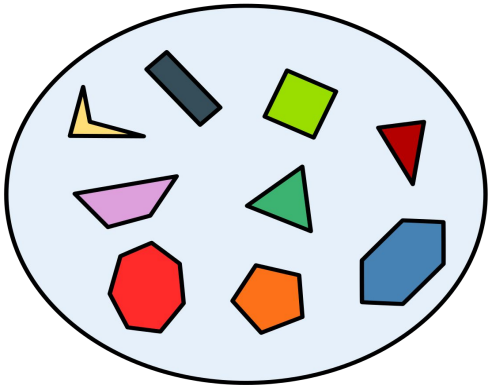
- "An *unordered ,well-defined*, collection of *distinct* objects (Called elements or members of a set) of same type". Here the type is defined by the one who is defining the set. For e.g.

- A = {0,2,4,6, ---}

- B = {1,3,5, ---}

- C= {x| x ∈ Natural number}

- A Set is generally denoted usually by capital letter. The objects of a set called the elements, or members of the set.
- A set is said to contain its elements. Lower case letters are generally used to denote the elements of the set.
- $x \in A$, means element x is a member of A, $x \notin A$ means x is not a member of A





Representation of set

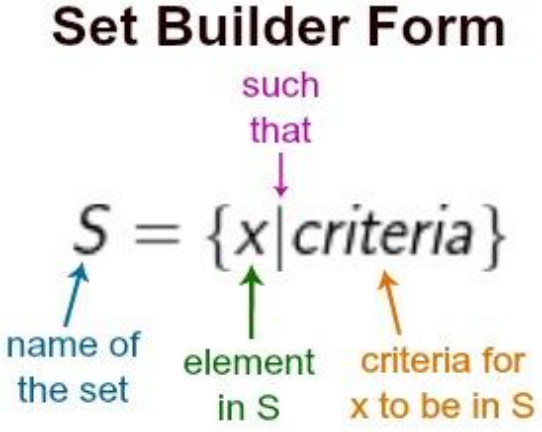
- **Tabular/Roster representation of set** - here a set is defined by actually listing its members. E.g.

- A = {a, e, i, o, u}

- B = {1, 2, 3, 4}

- C = {---- -4, -2, 0, 2, 4, -----}.

- **<u>Set Builder representations of set</u>**- here we specify the property which the elements of the set must satisfy. E.g.



- A = {x| x is an odd positive number less than 10},
- A = {x | x ϵ English alphabet && x is vowel}
- B = {x | x ϵ N && x <5}
- C = {x | x ϵ Z && x%2 = 0 }

- **<u>Finite set</u>** - If there are exactly '*n*' elements in *S* where '*n*' is a nonnegative integer, we say that *S* is a *finite set.*

- *i.e. if a* set contain specific or finite number of elements is called is called finite set. For e.g. A = {1,2,3,4}

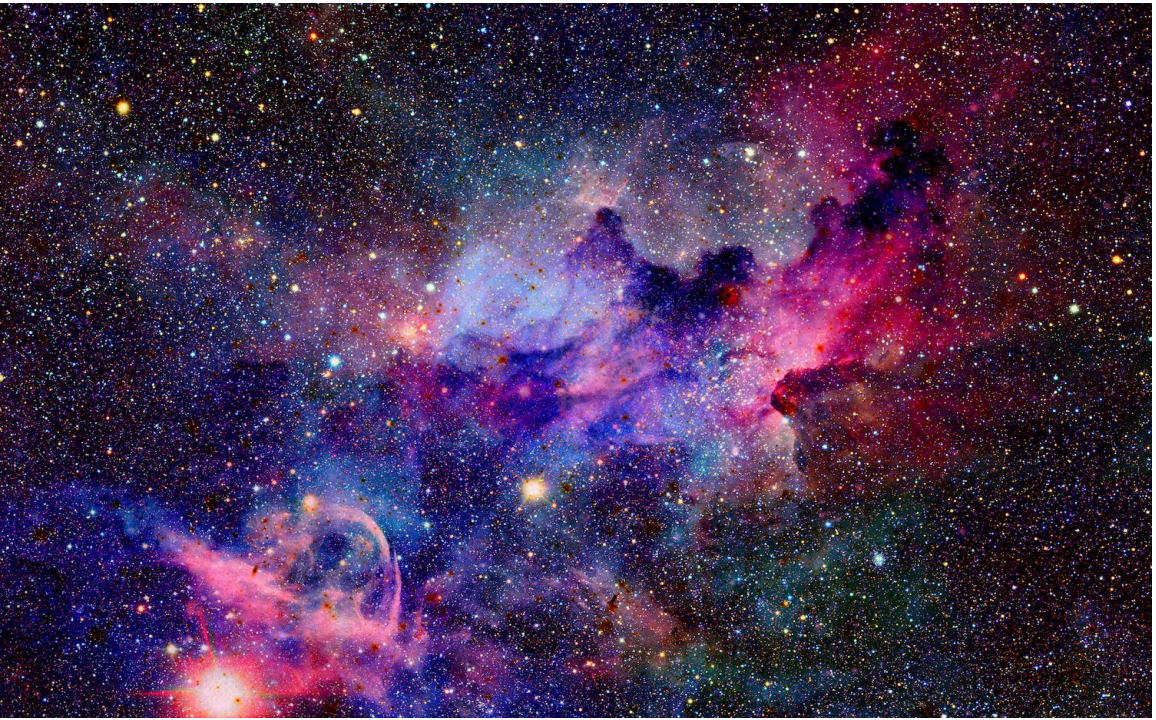
- **Cardinality of a set** — It is the number of elements present in a finite Set, denoted like |A|.

- For e.g. A = {0,2,4,6}, |A| = 4

- **Infinite set** – A set contain infinite number of elements is called infinite set, if the counting of different elements of the set does not come to an end. For e.g. a set of natural numbers.

- **<u>Null set / empty set</u>** - Is the unique set having no elements. its size or cardinality is zero i.e. $|\phi| = 0$. It is denoted by a symbol ϕ or {}.

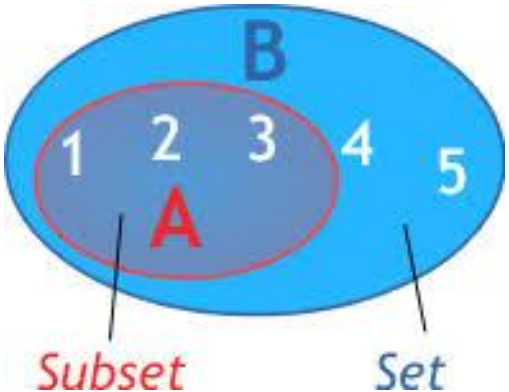
- **<u>Universal set</u>** – if all the sets under investigation are subsets of a fixed set, i.e. the set containing all objects under investigation, in Venn diagram it is represented by a rectangle, and it is denoted by U.



- **<u>Subset of a set</u>** – If every element of set A is also an element of set B i.e.
- $\forall x(x \in A \rightarrow x \in B)$, then A is called subset of B and is written as $A \sqsubseteq B$. B is called the superset of A.

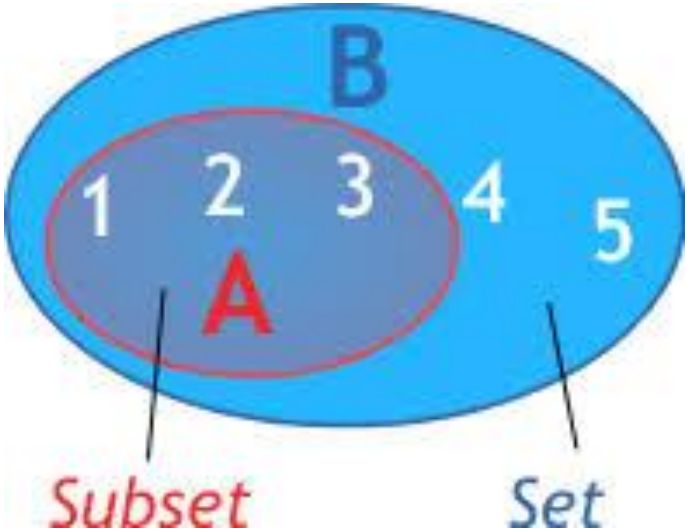
- E.g. A = {1,2,3} B = {1,2,3,4,5}

- Note that to show that *A* is not a subset of *B* we need only find one element $x \in A$ with $x \notin B$. To show that $A \sqsubseteq B$, show that if $x \in A$, then $x \in B$.

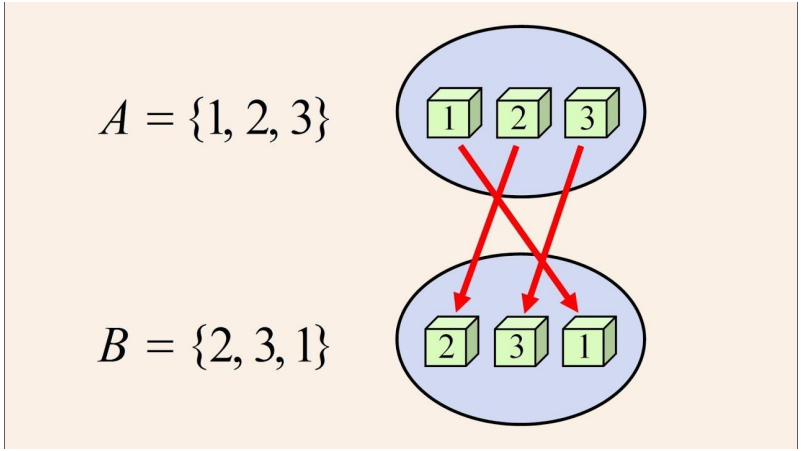


- $\phi \sqsubseteq A$, Empty Set ϕ is a subset for every set
- $A \sqsubseteq U$, Every Set is a subset of Universal set U
- $A \sqsubseteq A$, Every Set is a subset of itself.

- **<u>Proper subset</u> –** if A is a subset of B and A ≠ B, then A is said to be a proper subset of B, i.e. there is at least one element in B which is not in A. denoted as A ⊂ B.



- **<u>Equality of sets</u>** – If two sets A and B have the same element and therefore every element of A also belong to B and every element of B also belong to A, then the set A and B are said to be equal and written as A=B.
- if A $\sqsubseteq$ B and B $\sqsubseteq$ A, then A=B
- $\forall x(x \in A \leftrightarrow x \in B)$



- **Power set** – let A be any set, then the set of all subsets of A is called power set of A and it is denoted by P(A) or 2^A.

- If A= {1,2,3}, then P(A) = {ϕ, {1}, {2}, {3}, {1,2}, {2,3}, {1,3}, {1,2,3}}

- Cardinality of the power set of A is n, $|P(A)| = 2^n$

Q For any Set A, which of the following are true?

1) $\phi \in A$

2) $\phi \sqsubseteq A$

3) $\phi \in 2^A$

4) $\phi \sqsubseteq 2^A$

5) $A \in 2^A$

6) $A \sqsubseteq 2^A$

Q If ϕ is an empty set. Then | P(P(P(ϕ))) | =_____?

a) 1 **b)** 2 **c)** 4 **d)** none of above

Q The cardinality of the power set of {0, 1, 2 . . ., 10} is _____.

(GATE-2015) (1 Marks)

(A) 1024 **(B)** 1023 **(C)** 2048 **(D)** 2043

Q. The number of elements in the power set of { {1, 2}, {2, 1, 1}, {2, 1, 1, 2} } is: **(Math)(ISRO-2017-DEC)**

(A) 3
(B) 8
(C) 4
(D) 2

Q. The number of elements in the power set of the set {{A,B},C} is **(Math)(ISRO-2013)**

(A) 7
(B) 8
(C) 3
(D) 4

Q. The power set of the set {ϕ} is: **(NET-Dec-2012)**

a) {ϕ}

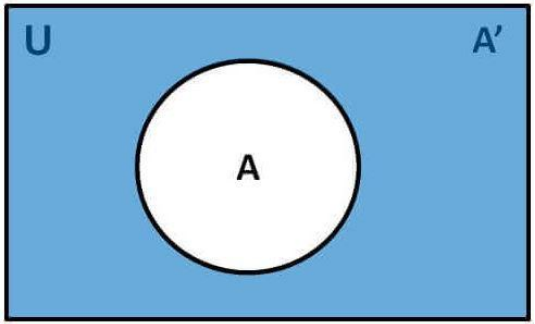
b) {ϕ, {ϕ}}

c) {0}

d) {0, ϕ, {ϕ}}

Operation on sets

- **Complement of set** – Set of all x such that $x \notin A$, but $x \in U$.
- $A^c = \{x \mid x \notin A \ \& \ x \in U\}$

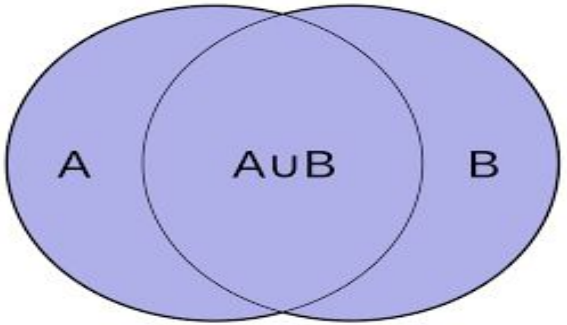


U = {1, 2, 3, 4, 5, 6}

A = {2, 3, 6}

A^c = { }

- **<u>Union of sets</u>** – Union of two sets A and B is a set of all those elements which either belong to A or B or both, it is denoted by A U B.
- A U B = {x| x ϵ A or x ϵ B}



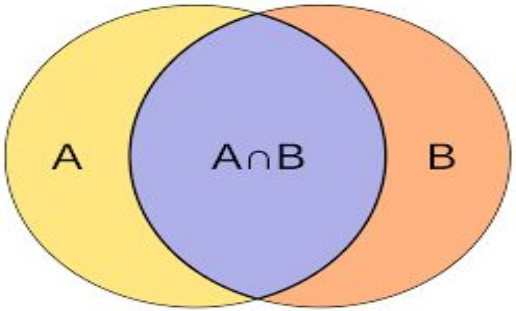
A = {1, 2, 3, 4}

B = {3, 4, 5, 6}

A U B = { }

|A|+|B|=| A U B | ?

- **<u>Intersection of sets</u>** - Intersection of two sets A and B is a set of all those elements which belong to both A and B, and is denoted by $A \cap B$.
- $A \cap B = \{x | \ x \in A \text{ and } x \in B\}$



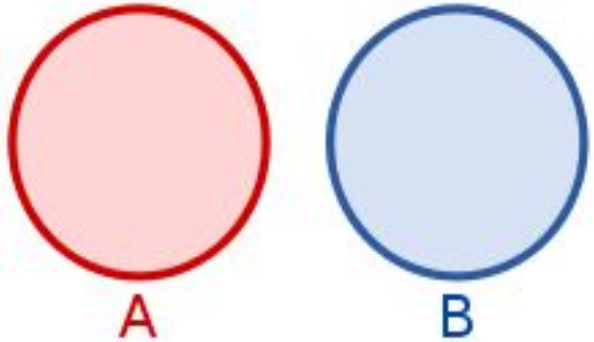
A = {1, 2, 3, 4}

B = {3, 4, 5, 6}

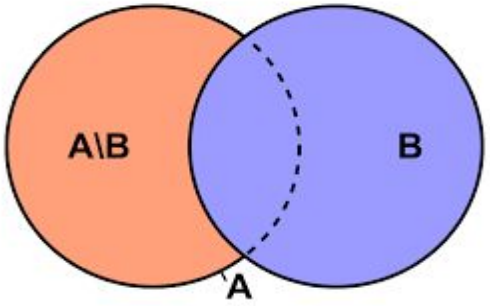
$A \cap B$ = { }

- **<u>Disjoint sets</u>** -- Two sets are said to be disjoint if they do not have a common element, i.e. no element in A is in B and no element in B is in A.

- $A \cap B = \phi$



- **<u>Set difference</u>** – the set difference of two sets A and B, is the set of all the elements which belongs to A but do not belong to B.
- $A - B = \{x | \ x \in A \text{ and } x \notin B\}$



$A = \{1, 2, 3, 4\}$

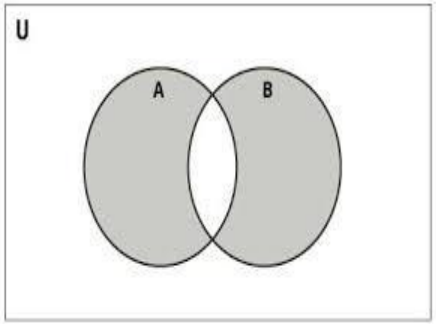
$B = \{3, 4, 5, 6\}$

$A - B = \{ \quad \}$

Q. Which one of the following is true? **(Math)(ISRO-2011)**

(A) $R \cap S = (R \cup S) - [(R - S) \cup (S - R)]$
(B) $R \cup S = (R \cap S) - [(R - S) \cup (S - R)]$
(C) $R \cap S = (R \cup S) - [(R - S) \cap (S - R)]$
(D) $R \cap S = (R \cup S) \cup (R - S)$

<u>Symmetric difference</u> – the symmetric difference of two sets A and B is the set of all the elements that are in A or in B but not in both, denoted as. $A \oplus B = (A \cup B) - (A \cap B)$



$A \oplus B = \{x \mid (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)\}$

$A \oplus B = (A - B) \cup (B - A)$

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

$A \oplus B = \{ \quad \}$

Q If A = {x, y, z} and B = {u, v, w, x}, and the universe is {s, t, u, v, w, x, y, z}. Then (A U B') ∩ (A ∩ B) is equal to: **(Math)(ISRO-2020)**

(A) {u, v, w, x}
(B) { }
(C) {u, v, w, x, y, z}
(D) {u, v, w}
(E) None of these

Idempotent law

- $A \cup A = A$
- $A \cap A = A$

Associative law

- $(A \cup B) \cup C = A \cup (B \cup C)$
- $(A \cap B) \cap C = A \cap (B \cap C)$

Commutative law

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

Distributive law

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

De Morgan's law

- $(A \cup B)^C = A^C \cap B^C$
- $(A \cap B)^C = A^C \cup B^C$

Identity law

- $A \cup \phi = A$
- $A \cap \phi = \phi$
- $A \cup U = U$
- $A \cap U = A$

Complement law

- $A \cup A^C = U$
- $A \cap A^C = \phi$
- $U^C = \phi$
- $\phi^C = U$

Involution law

- $((A)^C)^C = A$

Q which of the following is not true?

a) $A - B = A \cap B^c$

b) $A - (A - B) = A \cap B$

c) $A - (A \cap B) = A - B$

d) $A - (A - B) = B$

Q If $A \subset B$, then which of the following is not true?

(a) $A \cup B = B$

(b) $A \cap B = A$

(c) $B^C \subset A^C$

(d) $B - A = \phi$

Q Which the following in not true?

a) If $A \sqsubseteq \phi$, then $A = \phi$

b) $(A \cap B^C) \cup (A \cap B) = A$

c) $B \cup (A \cap B) = B$

d) $(A \cap B^C) \cup (A \cap B) \cup (A^C \cap B) \cup (A^C \cap B^C) = A \cap B$

Q Which of the following is true?

(i) $(A - B) - C = A - (C - B)$

(ii) $(A - B) - C = (A - C) - B$

(iii) $(A - B) - C = A - (B \cap C)$

(iv) $(A \cap B) - (B \cap C) = \{A - (A \cap C)\} - (A - B)$

a) i & iii **b)** ii & iv

c) i, ii, iv **d)** ii & iii

Q. The symmetric difference of sets A = {1, 2, 3, 4, 5, 6, 7, 8} and B = {1, 3, 5, 6, 7, 8, 9} is: **(Math)(ISRO-2017)**

(A) {1, 3, 5, 6, 7, 8}
(B) {2, 4, 9}
(C) {2, 4}
(D) {1, 2, 3, 4, 5, 6, 7, 8, 9}

Q. How many multiples of 6 are there between the following pairs of numbers? **(NET-Jan-2017)**

0 and 100 and –6 and 34

a) 16 and 6

b) 17 and 6

c) 17 and 7

d) 16 and 7

Q. Let A and B be sets in a finite universal set U. Given the following:

$|A - B|$, $|A \oplus B|$, $|A| + |B|$, and $|A \cup B|$

Which of the following is in order of increasing size? **(NET-Dec-2016)**

a) $|A - B| < |A \oplus B| < |A| + |B| < |A \cup B|$

b) $|A \oplus B| < |A - B| < |A \cup B| < |A| + |B|$

c) $|A \oplus B| < |A| + |B| < |A - B| < |A \cup B|$

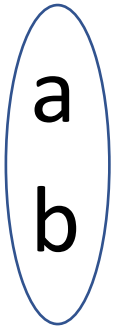
d) $|A - B| < |A \oplus B| < |A \cup B| < |A| + |B|$

Q. The power set of A ∪ B, where A = {2, 3, 5, 7} and B = {2, 5, 8, 9} is:
(NET-Dec-2012)

a) 256
b) 64
c) 16
d) 4

Cartesian Product

- Cartesian Product of two sets A and B in the set of all ordered pairs, whose first member belongs to the first set and second member belongs to the second set, denoted by $A \times B$. It is a kind of maximum relation possible, where every member of the first set belong to every member of the second set.
- $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$
- For E.g. if A = {a, b}, B = {1, 2, 3}
- A×B = { }





1. In general, commutative law does not hold good A× B != B×A

2. If |A| = m and |B| = n then |A×B| =

Relation

- **Relation**: - Let A and B are sets then every possible subset of ‘A×B’ is called a relation from A to B.

- If |A| =m and |B| = n then total no of element(pair) will be m*n, every element will have two choice weather to present or not present in the subset(relation), therefore the total number of relation possible is _____



For E.g. if A = {a, b}, B = {1, 2}, A×B = {(a, 1), (a, 2), (b, 1), (b, 2)}

(a , 1)	(a , 2)	(b , 1)	(b , 2)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, 2)}
0	0	1	0	2	{(b, 1)}
0	0	1	1	3	{(b, 1), (b, 2)}
0	1	0	0	4	{(a, 2)}
0	1	0	1	5	{(a, 2), (b, 2)}
0	1	1	0	6	{(a, 2), (b, 1)}
0	1	1	1	7	{(a, 2), (b, 1), (b, 2)}
1	0	0	0	8	{(a, 1)}
1	0	0	1	9	{(a, 1), (b, 2)}
1	0	1	0	10	{(a, 1), (b, 1)}
1	0	1	1	11	{(a, 1), (b, 1), (b, 2)}
1	1	0	0	12	{(a, 1), (a, 2)}
1	1	0	1	13	{(a, 1), (a, 2), (b, 2)}
1	1	1	0	14	{(a, 1), (a, 2), (b, 1)}
1	1	1	1	15	{(a, 1), (a, 2), (b, 1), (b, 2)}

- Largest relation possible will be _____
- Smallest possible relation will be _____

- **<u>Complement of a relation</u>**: - Let R be a relation from A to B, then the complement of relation will be denoted by R', R^C or R.
- R' = {(a, b)|(a, b) ϵ A×B, (a, b) ϵ! R}
- R' = (A×B) – R
- For E.g. if A×B = {(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)}
- R= {(a, 1), (a, 3), (b, 2)}
- R' = { }
- R U R' =
- R ∩ R' =

- **<u>Inverse of a relation</u>**: - Let R be a relation from A to B, then the inverse of relation will be a relation from B to A, denoted by R^{-1}.

- $R^{-1} = \{(b, a) \mid (a, b) \in R\}$

- $A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$

- $R = \{(a, 1), (a, 3), (b, 2)\}$

- $R^{-1} = \{$ $\}$

- $|R|$ $|R^{-1}|$

Q The number of binary relations on a set with n elements is: **(GATE-1999) (1 Marks)**

(A) n^2

(B) 2^n

(C) 2^n^2

(D) None of the above

Q Let A be a finite set of size n. The number of elements in the power set of A × A is: **(GATE-1993) (1 Marks)**

a) $2^{(2^n)}$

b) $2^{(n^2)}$

c) 2^n

d) None of the above

- **<u>Diagonal relation</u>**: - A relation R on a set A is said to be diagonal relation if, R is a set of all ordered pair (x, x), for every $\forall x \in A$, sometimes it is also denoted by $\blacktriangle_A$

- $R = \{(x, x) \mid \forall\ x \in A\}$

	1	**2**	**3**
1	11		
2		22	
3			33

Types of a Relation

- To further study types of relations, we consider a set A with n elements, then a cartesian product A×A will have n^2 elements(pairs). Therefore, total number of relation possible is $2^{n * n}$

- **Reflexive relation**: - A relation R on a set A is said to be reflexive,
- If $\forall x \in A$
  - $(x, x) \in R$

	1	2	3
1	11		
2		22	
3			33

Q consider a set A = {1,2,3}, find which of the following relations are reflexive and Irreflexive?

	Relation	Reflexive	Irreflexive
1	A×A		
2	ϕ		
3	{(1,1), (2,2), (3,3)}		
4	{(1,2), (2,3), (1,3)}		
5	{(1,1), (1,2), (2,1), (2,2)}		
6	{(1,1), (2,2), (3,3), (1,3), (2,1)}		
7	{(1,3), (2,1), (2,3), (3,2)}		

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	
0	0	0	1	1	
0	0	1	0	2	
0	0	1	1	3	
0	1	0	0	4	
0	1	0	1	5	
0	1	1	0	6	
0	1	1	1	7	
1	0	0	0	8	
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

Q What is the possible number of reflexive relations on a set of 5 elements? **(GATE-2010) (UP-LT 2018) (1 Marks)**

(A) 2^{10} **(B)** 2^{15} **(C)** 2^{20} **(D)** 2^{25}

Q How many reflexive relations are there on a set with 4 elements? **(NET 2019 DEC)**

(A) 2^4

(B) 2^{12}

(C) 4^2

(D) 2

- **<u>Irreflexive relation</u>**: - A relation R on a set A is said to be Irreflexive,

1. If $\forall\ x \in A$
2. $(x, x) \notin R$

Q consider a set A = {1,2,3}, find which of the following relations are reflexive and Irreflexive?

	Relation	Reflexive	Irreflexive
1	A×A	Y	
2	ϕ	N	
3	{(1,1), (2,2), (3,3)}	Y	
4	{(1,2), (2,3), (1,3)}	N	
5	{(1,1), (1,2), (2,1), (2,2)}	N	
6	{(1,1), (2,2), (3,3), (1,3), (2,1)}	Y	
7	{(1,3), (2,1), (2,3), (3,2)}	N	

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	
0	0	1	0	2	{(b, a)}
0	0	1	1	3	
0	1	0	0	4	{(a, b)}
0	1	0	1	5	
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	
1	0	0	0	8	
1	0	0	1	9	
1	0	1	0	10	
1	0	1	1	11	
1	1	0	0	12	
1	1	0	1	13	
1	1	1	0	14	
1	1	1	1	15	

- **<u>Symmetric relation</u>**: - A relation R on a set A is said to be Symmetric,

If $\forall$ a, b $\in$ A

(a, b) $\in$ R

.....

then (b, a) $\in$ R

.....

Q Consider a set A = {1,2,3}, find which of the following relations are Symmetric, Anti-Symmetric and Asymmetric?

	Relation	Symmetric	Anti-Symmetric	Asymmetric
1	A×A			
2	ϕ			
3	{(1,1), (2,2), (3,3)}			
4	{(1,2), (2,3), (1,3)}			
5	{(1,1), (1,2), (2,1), (2,2)}			
6	{(1,1), (2,2), (3,3), (1,3), (2,1)}			
7	{(1,3), (2,1), (2,3), (3,2)}			

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	
0	0	1	1	3	
0	1	0	0	4	
0	1	0	1	5	
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	
1	0	1	1	11	
1	1	0	0	12	
1	1	0	1	13	
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

- **<u>Anti-Symmetric relation</u>**: - A relation R on a set A with cartesian product A×A is said to be Anti-Symmetric,

If $\forall$ a, b $\in$ A

(a, b) $\in$ R

(b, a) $\in$ R

……

a = b

……

Conclusion: Symmetry is not allowed but diagonal pairs are allowed

	Relation	Symmetric	Anti-Symmetric	Asymmetric
1	$A \times A$	Y		
2	ϕ	Y		
3	{(1,1), (2,2), (3,3)}	Y		
4	{(1,2), (2,3), (1,3)}	N		
5	{(1,1), (1,2), (2,1), (2,2)}	Y		
6	{(1,1), (2,2), (3,3), (1,3), (2,1)}	N		
7	{(1,3), (2,1), (2,3), (3,2)}	N		

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	
0	1	1	1	7	
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	
1	1	1	1	15	

Q Consider a set A = {a, b, c} and R_1, R_2, R_3 and R_4 are relations on A which of the following is not true?

		Symmetric	Anti-Symmetric	True
1	$R_1 = \{(a, a), (c, c)\}$	Y	Y	
2	$R_2 = \{(a, b), (b, a), (a, c)\}$	N	N	
3	$R_3 = \{(a, b), (b, c), (a, c)\}$	N	Y	
4	$R_4 = \{(a, b), (b, a), (c, c)\}$	Y	N	

Q Consider the binary relation R = {(x, y), (x, z), (z, x), (z, y)} on the set {x, y, z}. Which one of the following is TRUE? **(GATE-2009) (1 Marks)**

(A) R is symmetric but NOT antisymmetric

(B) R is NOT symmetric but antisymmetric

(C) R is both symmetric and antisymmetric

(D) R is neither symmetric nor antisymmetric

- **Asymmetric relation**: - A relation R on a set A is said to be Asymmetric,

If $\forall$ a, b $\in$ A

(a, b) $\in$ R

……

(b, a) $\notin$ R

……

Conclusion: Symmetry is not allowed; even diagonal pairs are not allowed

	Relation	Symmetric	Anti-Symmetric	Asymmetric
1	A×A	Y	N	
2	ϕ	Y	Y	
3	{(1,1), (2,2), (3,3)}	Y	Y	
4	{(1,2), (2,3), (1,3)}	N	Y	
5	{(1,1), (1,2), (2,1), (2,2)}	Y	N	
6	{(1,1), (2,2), (3,3), (1,3), (2,1)}	N	Y	
7	{(1,3), (2,1), (2,3), (3,2)}	N	N	

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	
0	0	1	0	2	{(b, a)}
0	0	1	1	3	
0	1	0	0	4	{(a, b)}
0	1	0	1	5	
0	1	1	0	6	
0	1	1	1	7	
1	0	0	0	8	
1	0	0	1	9	
1	0	1	0	10	
1	0	1	1	11	
1	1	0	0	12	
1	1	0	1	13	
1	1	1	0	14	
1	1	1	1	15	

- **<u>Transitive relation</u>**: - A relation R on a set A is said to be Transitive,

If $\forall$ a, b, c $\in$ A

(a, b) $\in$ R

(b, c) $\in$ R

.....

(a, c) $\in$ R

.....

	Relation	Transitive
1	$A \times A$	
2	ϕ	
3	{(1,1), (2,2), (3,3)}	
4	{(1,2), (2,3), (1,3)}	
5	{(1,1), (1,2), (2,1), (2,2)}	
6	{(1,1), (2,2), (3,3), (1,3), (2,1)}	
7	{(1,3), (2,1), (2,3), (3,2)}	
8	{(1,2)}	
9	{(1,3), (2,3)}	
10	{(1,2), (1,3)}	
11	{(2,3), (1,2)}	

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	{(a, b), (b, a)}
0	1	1	1	7	{(a, b), (b, a), (b, b)}
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	{(a, a), (a, b), (b, a)}
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

For E.g. if A = {a, b}, A×A = {(a, a), (a, b), (b, a), (b, b)}

(a , a)	(a , b)	(b , a)	(b , b)		
0	0	0	0	0	{ }
0	0	0	1	1	{(b, b)}
0	0	1	0	2	{(b, a)}
0	0	1	1	3	{(b, a), (b, b)}
0	1	0	0	4	{(a, b)}
0	1	0	1	5	{(a, b), (b, b)}
0	1	1	0	6	
0	1	1	1	7	
1	0	0	0	8	{(a, a)}
1	0	0	1	9	{(a, a), (b, b)}
1	0	1	0	10	{(a, a), (b, a)}
1	0	1	1	11	{(a, a), (b, a), (b, b)}
1	1	0	0	12	{(a, a), (a, b)}
1	1	0	1	13	{(a, a), (a, b), (b, b)}
1	1	1	0	14	
1	1	1	1	15	{(a, a), (a, b), (b, a), (b, b)}

Q Consider the following properties:

a. Reflexive
b. Antisymmetric
c. Symmetric

Let A={a,b,c,d,e,f,g} and R={(a,a),(b,b),(c,d),(c,g),(d,g),(e,e),(f,f),(g,g)} be a relation on A. Which of the following property (properties) is (are) satisfied by the relation R?**(NET 2020 OCT)**

(A) Only a

(B) Only c

(C) Both a and b

(D) b and not a

Q. A relation **R** in {1, 2, 3, 4, 5, 6} is given by {(1,2), (2,3), (3,4), (4,4), (4,5)}. This relation is: **(NET-Dec-2008)**

a) Reflexive

b) Symmetric

c) Transitive

d) not reflexive, not symmetric and not transitive

- **<u>Equivalence Relation</u>**: - A relation R on a set A with cartesian product A×A is said to be Equivalence, if it is

1. **Reflexive**
2. **Symmetric**
3. **Transitive**

- If two relations R_1 and R_2 are Equivalence then their union need not to be equivalence but intersection will also be Equivalence.

Q. In the relation R={(1,2),(2,3)}, the minimum number of ordered pairs that must be added to this set, so that the enlarged relation is reflexive, symmetric and transitive, is **(UP-LT 2018)**

(a) 4

(b) 5

(c) 6

(d) 7

Q. The number of equivalence relations of the set {1,2,3,4} is **(UP-LT 2018)**

(a) 4

(b) 24

(c) 25

(d) 15

Q Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are **(GATE-2007) (1 Marks)**

(A) n and n

(B) n^2 and n

(C) n^2 and 0

(D) n and 1

Q Which of the relations on {0, 1, 2, 3} is an equivalence relation? **(NET-July-2018)**

a) {(0, 0) (0, 2) (2, 0) (2, 2) (2, 3) (3, 2) (3, 3)}

b) {(0, 0) (1, 1) (2, 2) (3, 3)}

c) {(0, 0) (0, 1) (0, 2) (1, 0) (1, 1) (1, 2) (2, 0)}

d) {(0, 0) (0, 2) (2, 3) (1, 1) (2, 2)}

- **<u>Partial Order Relation</u>**: - A relation R on a set A with cartesian product A×A is said to be partial order, if it is

1. **Reflexive**
2. **Anti - Symmetric**
3. **Transitive**

- **<u>Partial ordering set (Poset):</u>** - a set A with partial ordering relation R defined on A is called a POSET and is denoted by [A, R]

- For e.g. [A, /], [A, <=], [P(S), $\sqsubseteq$]

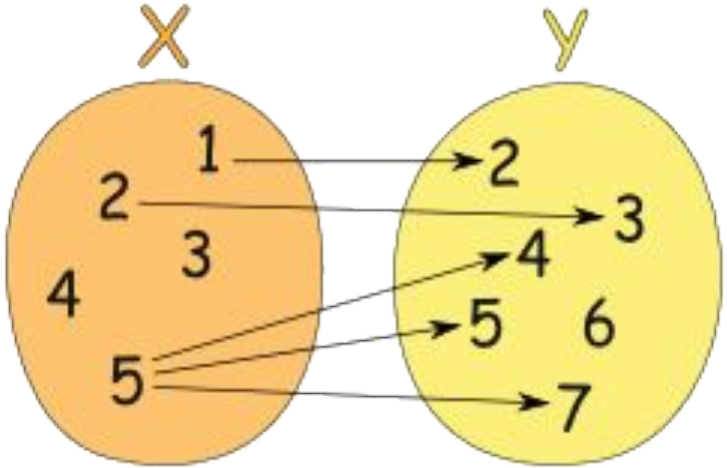
- **<u>Total order relation:</u>** - A Poset [A, R] is called a total order set, if every pair of elements are comparable i.e. either (a, b) or (b, a) $\in$ R, for $\forall$ a, b $\in$ A

- For e.g. A = {1, 2, 3, 6}, then Poset [A,/] is not a total order relation but A = {1,2,4,8} will be

Function

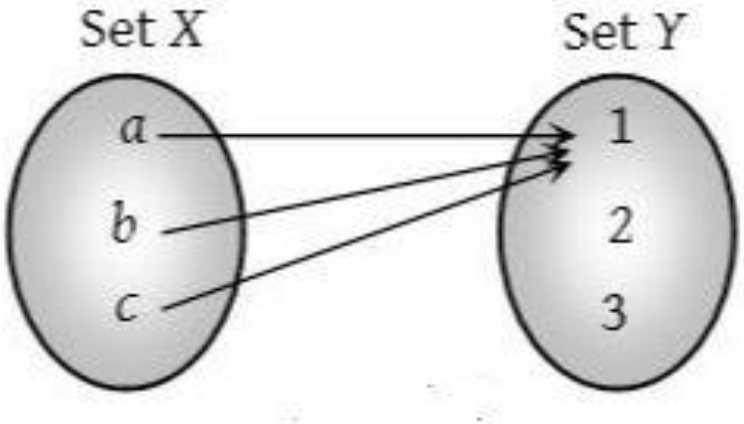
- Functions are widely used in science, and in most fields of mathematics. It has been said that functions are "the central objects of investigation" in most fields of mathematics.

- Let X = {1, 2, 3, 4, 5} and Y = {2, 3, 4, 5, 6, 7} and R ⊑ X * Y. Now this is a valid relation but not a function, because there is a element which is not participating in the relationship secondly 5 is relating with more than one element.



Function

- In mathematics, a function is a relation between sets that associates to every element of a first set exactly one element of the second set.
- A relation ‘f’ from a set ‘A’ to a Set ‘B’ is called a function, if each element of A is mapped with a unique element on B.
- f: A□B



- Range of fun ⊑ B
- Range of f = { y | y ϵ B and (x, y) ϵ f}

- If |A| = m and |B|= n, then number of functions possible from A to B = ?

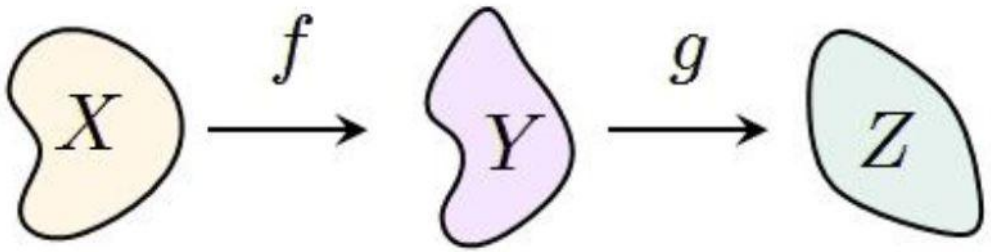
Q The number of functions from an m element set to an n element set is **(GATE-1998) (1 Marks)**

a) $m + n$ **b)** m^n **c)** n^m **d)** $m * n$

Function composition

- In mathematics, **function composition** is an operation that takes two functions *f* and *g* and produces a function *h* such that $h(x) = g(f(x))$.

- In this operation, the function *g* is applied to the result of applying the function *f* to *x*. That is, the functions $f: X \to Y$ and $g: Y \to Z$ are **composed** to yield a function that maps *x* in *X* to $g(f(x))$ in *Z*.



- fog(x) = f(g(x))
- gof(x) = g(f(x))

- Composition of functions on a finite set: If $f = \{(1, 3), (2, 1), (3, 4), (4, 6)\}$, and $g = \{(1, 5), (2, 3), (3, 4), (4, 1), (5, 3), (6, 2)\}$, then $g \circ f = \{(1, 4), (2, 5), (3, 1), (4, 2)\}$.
- The composition of functions is always associative—a property inherited from the composition of relations. That is, if f, g, and h are three functions with suitably chosen domains and codomains, then $f \circ (g \circ h) = (f \circ g) \circ h$

Q If $g(x) = 1-x$ and $h(x)=x / (x-1)$, then $g(h(x)) / h(g(x))$ is: **(GATE-2015) (1 Marks)**

a) $h(x) / g(x)$ **b)** $-1 / x$ **c)** $g(x) / h(x)$ **d)** $x / (1-x)^2$

Q. Let f and g be the functions from the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. Then the composition of f and g and g and f is given as

(NET-Jan-2013)

a) $6x + 7, 6x + 11$

b) $6x + 11, 6x + 7$

c) $5x + 5, 5x + 5$

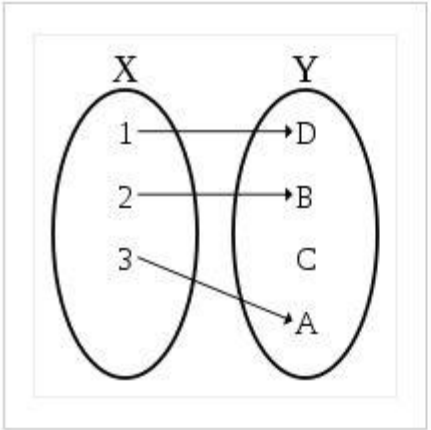
d) None of the above

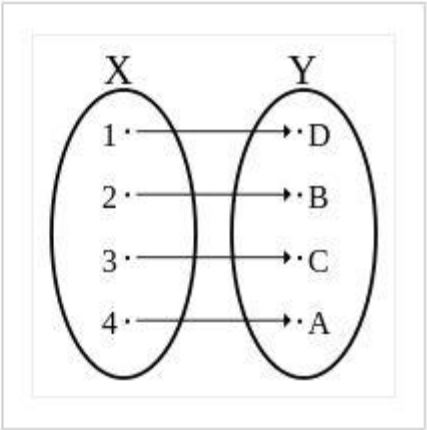
One-to-One (Injective Function)

- An **injective function** (also known as **injection**, or **one-to-one function**) is a function that maps distinct elements of its domain to distinct elements of its codomain. In other words, every element of the function's codomain is the image of *at most* one element of its domain.

One-to-One (Injective Function)

- A function F: A□B is said to be one-to-one function if every element of A has distinct image in B
- If A and B are finite set, then one-to-one from A□B is possible
  - if |A| <= |B|





- No of function possible = ?

Q Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X and Y. Let f be randomly chosen from F. The probability of f being one-to-one is _____ **(GATE-2015) (2 Marks)**

Q Let A and B be non-empty finite sets such that there exists one-to-one and onto functions

(i) from A to B

(ii) from AxA to A U B.

The number of possible values of |A| is _____ **(Gate-2024,CS)(1 Marks)(NAT)**

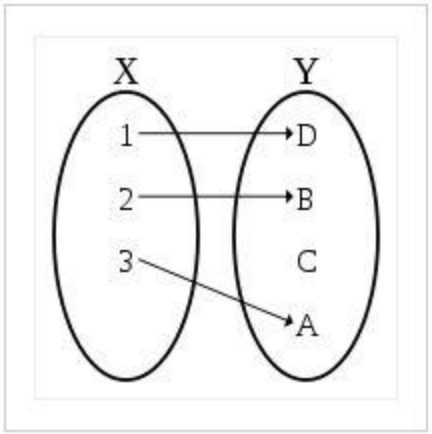
- No of function possible = ${}^{n}p_{m}$ = P (n, m)
- If |A| = |B| = n, then no of functions possible is n!

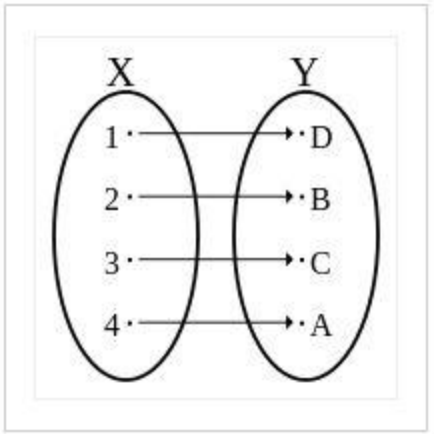
Onto (Surjective Function)

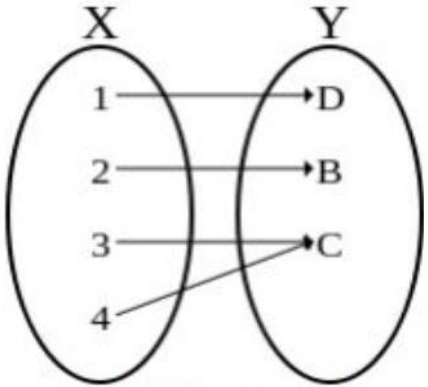
- A function f from a set X to a set Y is **surjective** (also known as **onto**, or a **surjection**), if for every element y in the co-domain Y of f, there is at least one element x in the domain X of f such that $f(x) = y$. It is not required that x be unique; the function f may map one or more elements of X to the same element of Y.

Onto (Surjective Function)

- A function f: A □ B is said to be onto if and only if every element of B is mapped by at least one element of A.
- Range of f = B







- If A and B are finite sets, then onto function from A□B is possible, |B|<=|A|
- If |A| = |B|, then every onto function from A to B is also one-to-one function.

No of onto function possible from A to B = ?

No of onto function possible from A to B

$= n^m - {}^nc_1(n-1)^m + {}^nc_2(n-2)^m - {}^nc_3(n-3)^m + \text{---------} + (-1)^n\ {}^nc_{n-1}\ 1^m$

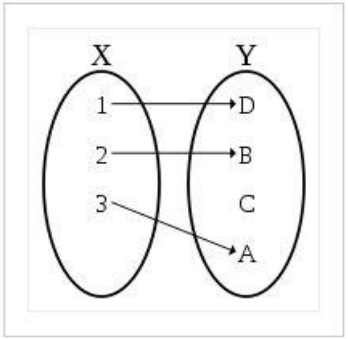
Q The number of onto functions (surjective functions) from set X = {1, 2, 3, 4} to set Y = {a, b, c} is _____ **(GATE-2015) (2 Marks)**

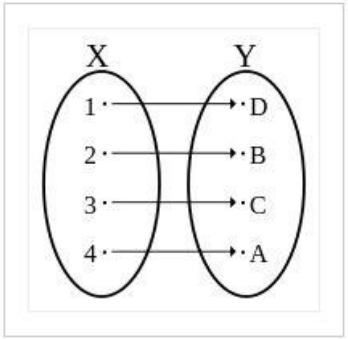
Q How many onto (or surjective) functions are there from an n-element (n >= 2) set to a 2-element set? **(GATE-2014) (2 Marks)**

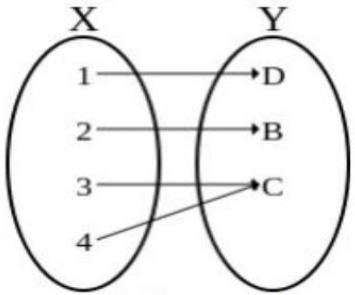
(A) 2^n **(B)** $2^n - 1$ **(C)** $2^n - 2$ **(D)** $2(2^n - 2)$

Bijective Function

- In mathematics, a **bijection**, **bijective function**, **one-to-one correspondence**, or **invertible function**, is a function between the elements of two sets, where each element of one set is paired with exactly one element of the other set, and each element of the other set is paired with exactly one element of the first set. There are no unpaired elements.







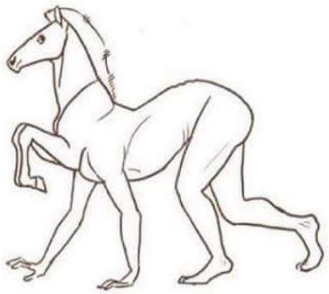
- A function f: A□B is said to be bijection if f is one-to-one and onto.
- Bijection from A and B is possible, if□|A| |B|
- No of Bijection from A to B = ?

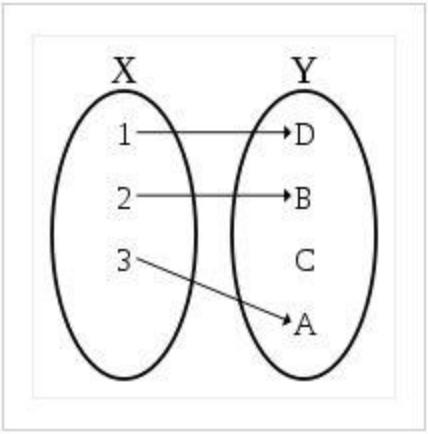
- A function f: A□B is said to be bijection if f is one-to-one and onto.
- Bijection from A and B is possible, if |A| = |B|
- No of Bijection from A to B = n!

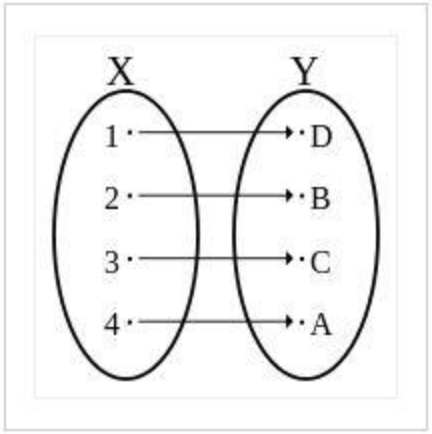
Inverse of a function

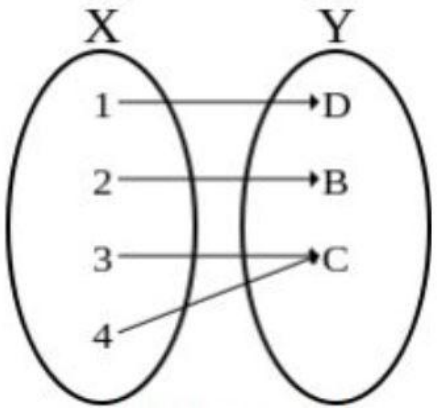
- In mathematics, an **inverse function** (or **anti-function**) is a function that "reverses" another function
- If the function *f* applied to an input *x* gives a result of *y*, then applying its inverse function f^{-1} to *y* gives the result *x*, and vice versa.
- $f(x) = y$ then $f^{-1}(y) = x$.











- Inverse of a function f: A->B exists, iff f: A->B is a bijection.

- $f(x) = 5x - 7$
- $f^{-1}(y) = (y + 7)/5$